Infectious Diseases Now 53 (2023) 104628

Available online at
ScienceDirect
www.sciencedirect.com

Elsevier Masson France
EM|consulte
www.em-consulte.com/en

Review

The crisis of carbapenemase-mediated carbapenem resistance across the human–animal–environmental interface in India

Surojit Das

Biomedical Laboratory Science and Management, Vidyasagar University, Midnapore 721102, West Bengal, India

ARTICLE INFO

Article history:
Received 7 July 2022
Revised 11 September 2022
Accepted 30 September 2022
Available online 12 October 2022

Keywords:
Carbapenem-resistance
Carbapenemases
Plasmid
Clones
One health
India

ABSTRACT

Carbapenems are the decision-making antimicrobials used to combat severe Gram-negative bacterial infections in humans. Carbapenem resistance poses a potential public health emergency, especially in developing countries such as India, accounting for high morbidity, mortality, and healthcare cost. Emergence and transmission of plasmid-mediated "*big five*" carbapenemase genes including KPC, NDM, IMP, VIM and OXA-48-type among Gram-negative bacteria is spiralling the issue. Carbapenemase-producing carbapenem-resistant organisms (CP-CRO) cause multi- or pan-drug resistance by co-harboring several antibiotic resistance determinants. In addition of human origin, animals and even environmental sites are also the reservoir of CROs. Spillage in food-chains compromises food safety and security and increases the chance of cross-border transmission of these superbugs. Metallo-β-lactamases, mainly NDM-1 producing CROs, are commonly shared between human, animal and environmental interfaces worldwide, including in India. Antimicrobial resistance (AMR) surveillance using the One Health approach has been implemented in Europe, the United-Kingdom and the United-States to mitigate the crisis. This concept is still not implemented in most developing countries, including India, where the burden of antibiotic-resistant bacteria is high. Lack of AMR surveillance in animal and environmental sectors underestimates the cumulative burden of carbapenem resistance resulting in the silent spread of these superbugs. In-depth indiscriminate AMR surveillance focusing on carbapenem resistance is urgently required to develop and deploy effective national policies for preserving the efficacy of carbapenems as last-resort antibiotics in India. Tracking and mapping of international high-risk clones are pivotal for containing the global spread of CP-CRO.

1. Introduction

Over the last two decades, rising antimicrobial resistance (AMR) has emerged as a global public health issue threatening healthcare fraternities. The Global Research on AMR Project (GRAM) study in 2019 estimated 4.95 million deaths associated with bacterial AMR severely affecting South Asia, and the Review on AMR in 2016 projected as many as 10 million deaths annually by 2050 [1]. From 2000 to 2010, the global antibiotic use in clinical, agricultural, food, and aquaculture settings increased by 35%, of which 79% was attributed to Brazil, Russia, India, China, and South Africa (BRICS) [2]. From 2009 to 2015, the highest drug resistance index was noted in India (range, 71–83) indicating the lowest effectiveness of antibiotic therapy [3,4]. Carbapenems, belonging to the watch group of essential medicines listed by the Word Health Organization (WHO), serve as decision-making antimicrobials for treating severe Gram-negative bacterial infections in humans. Since the last decade, a significant increase (45%) in carbapenem use has led to the emergence and global dissemination of carbapenem-resistant organisms (CROs) raising mortality and healthcare costs [2,4–6]. Carbapenemase-producing (CP)-CRO mediated bloodstream infection (BSI), community-acquired pneumonia (CAP), hospital-acquired pneumonia (HAP), ventilator-associated pneumonia (VAP), complicated urinary tract infections (cUTIs), and complicated intra-abdominal infections (cIAIs) cause significant case fatality, mainly due to sepsis [5–7]. Evidence of CP-CROs has been commonly documented in Europe, Asia and South America, with substantial variability in carbapenemases on the continental, national, regional and even healthcare centers at local levels [8–11]. Notably, India is the epicenter of New Delhi metallo-β-lactamase-(NDM-1) and OXA-48-type-producing CRO [5,9,12]. CP-CROs, especially NDM-1 producers, are not only restricted to humans but have also been isolated from animals and the environment [10]. This phenomenon indicates the importance of the One Health approach for the surveillance and containment of CP-CROs. Since the last decade, the United-States and European countries have taken initiatives for the detection and control of CROs following public health measure recommendations [13,14]. Asian countries, especially India, account for high burden of CROs. The

E-mail address: esurojitdas@gmail.com

https://doi.org/10.1016/j.idnow.2022.09.023

AMR research in India is mainly focused on human studies (>50%) neglecting the animal, environmental, or one health study (<10%) which underestimates the cumulative burden of AMR [2–4,15]. Notably, no review has been undertaken on the emergence of CROs from the one health perspective in India. In the current review, the global emergence of CP-CROs along with knowledge of international high-risk clones (HRCs) circulating through human, animal and environment interfaces has been depicted focusing on the situation in India.

2. Carbapenem resistance

Carbapenems are the last-resort β-lactam antibiotics for the treatment of humans because of their relative safety and efficacy in resolving AmpC cephalosporinase and extended-spectrum β-lactamase (ESBL)-mediated multidrug-resistant (MDR) Gram-negative bacterial infections [2]. Since the last decade, the gradual rise in carbapenem resistance has posed a massive challenge for clinicians in treating community- and health care-associated infections (CAIs and HAIs) [6,8,12]. Carbapenem resistance is mediated by three mechanisms: carbapenemase production, active efflux pumps, and porin loss [6,8,9]. Carbapenem-resistant Enterobacterales (CRE) commonly produce carbapenemases as the sole mechanism of resistance without requiring additional accessory mechanisms. CP-CRO confers higher carbapenem resistance than non-CP-CRO. Plasmid-borne carbapenemase genes disseminate rapidly via horizontal gene-transfer to commensal bacteria [8,9]. Co-mobilization of other plasmid-mediated AMR genes by CRO mediates co-resistance to several antibiotics, especially fluoroquinolones, aminoglycosides, and colistin [16,17]. In 2017, the WHO categorized CRE, *Pseudomonas aeruginosa* (CRPsA), and *Acinetobacter baumannii* (CRAB) as critical pathogens, with mortality rates ranging from 6.6% to 20% [1,16]. In India, researchers reported a high prevalence of carbapenem resistance in *Acinetobacter baumannii* (>70%), followed by *Klebsiella pneumoniae* (>50%), *Pseudomonas aeruginosa* (>40%), and *Escherichia coli* (>10%) [4,15]. CROs have been associated with several outbreaks in healthcare settings; notably, CRAB and CRPsA disseminate rapidly due to improper infection control practices compared to CRE [6,7,17]. One study estimated a cost of about €1.1 million over 10 months for managing the CRE outbreak in the United-Kingdom [7]. Moreover, specific antibiotics are needed against different carbapenemases because of their diverse hydrolytic spectra [8]. The crisis of new and effective antibiotics demands the precise detection of carbapenemase genes following the prudent use of currently available treatment regimens [2,5,9].

Ecosystems serve as a hidden source of antibiotic-resistant organisms harboring transferable resistance genes due to the deposition of significant quantities of antibiotic residues from the pharmaceutical industry, livestock farms, and hospitals as well as the fecal waste of humans and animals [3,17,18]. In 2010, India was ranked the fifth largest consumer of antibiotics in food animals because of the high demand for animal protein projecting to increase >300% by 2030 [4]. In animals, the use of extended spectrum cephalosporins, fluoroquinolones, tetracyclines, polymyxins, aminoglycosides, sulfonamides and trimethoprim selects and disperses resistant populations of bacteria carrying transferable resistance genes [19,20]. Although carbapenems are not approved for animal use, sporadic emergence of CP-CRO carrying NDM-1 in animals and the environment from several countries, including India, gradually complicates AMR [10,11,21,22]. Several factors may promote the dissemination of carbapenem resistance across humans and animals, such as high population, inadequate sanitation, contaminated drinking water, tropical climate, and antibiotic abuse [3,6,23].

3. Versatility of carbapenemases

An enormous variety of carbapenemases have been documented as belonging to the Ambler classes A, B and D β-lactamases [6]. Class A, C and D β-lactamases are known as serine β-lactamases (SBLs) containing serine at their active site, whereas class B members are familiar as metallo-β-lactamases (MBLs) containing an active-site zinc. The Ambler class C (primarily AmpC) β-lactamases possess a low potential of carbapenem hydrolysis, but their overproduction may confer carbapenem resistance combined with diminished outer-membrane permeability and/or efflux pump over-expression [8,9]. Class A carbapenemases are chromosomally encoded like NmcA (not metallo enzyme carbapenemase A), SME (*Serratia marcescens* enzyme), IMI-1 (imipenem-hydrolyzing β-lactamase), SFC-1 (*Serratia fonticola* carbapenemase-1), and the others are carried by plasmids including KPC (*Klebsiella pneumoniae* carbapenemase), IMI (IMI-2 and IMI-3), and few derivatives of GES (Guiana extended spectrum) [5,9]. Common class B MBLs are NDM-1, IMP (imipenemase), VIM (Verona integron-encoded MBL), SPM (Sao Paulo MBL), GIM (German imipenemase) and SIM (Seoul imipenemase). The MBL genes are often incorporated in gene cassettes of different integron classes. These enzymes hydrolyze all clinically available β-lactams except aztreonam, but are inhibited by ethylenediaminetetraacetic acid (EDTA), a chelator of divalent cations including Zn^{2+}. The class D carbapenemases are of the OXA (oxacillinase) enzyme type mainly including the type of OXA-48, OXA-23, OXA-24, OXA-51 and OXA-58. They have a weak activity against carbapenems and are poorly inhibited by EDTA or clavulanic acid [9]. In terms of carbapenem hydrolysis and geographic distribution, clinically important carbapenemases including KPC, NDM, VIM, IMP and OXA-48 types among nosocomial pathogens are known as "*big five*" enzymes [5].

4. Prevalence of "*big five* carbapenemases

The one health initiative has been adapted to mitigate the caveat of AMR and to optimize antibiotic usage by holistic and coordinated approach towards the health of humankind, animals and ecosystems [14,17,24]. In addition to human origin, the occurrence of CRO in animal and environmental sectors requires the one health approach for tracking and mapping the transmission of these super-bugs. Human health remains the highest priority, while surveillance in animals and aquaculture is neglected for assessing the burden of AMR, including carbapenem resistance, especially in developing countries like India. Overall, the relative preface of these three domains in the development, dissemination and persistence of carbapenem resistance remains unclear. A prudent one health initiative covering these domains will address the evolution and transmission of carbapenem resistance.

4.1. Human origin

Since 1996, the first identification of KPC producer (KPC-2 in *K. pneumoniae*) in the United-States, it has now spread in almost all US states and become endemic to sporadic in other parts of the world including North America, Europe, and Southeast Asia as well as Africa [5,8]. More than 90 KPC variants have been described, among which KPC-2 and KPC-3 are commonly reported in different studies, including India [9,12]. KPC producers were sporadically reported from north and east India (Table 1) [25–28]. Rahman et al. detected a significant proportion (38%) of KPC in CPE mostly co-producing NDM-1 from Lucknow, north India [29]. On the contrary, KPC was not found in south India [12,30]. KPC has been commonly detected in nosocomial carbapenem-resistant *K.*

Table 1
Distribution of clinically important carbapenemases among human-derived isolates in India.

Region/Location	Period	Sample	CRO (*n*)	CP-CRO (*n*)	Carbapenemases (*n*)[a]					Ref.
					KPC	NDM	VIM	IMP	OXA-48-type	
North										
Haryana, Haryana	2009	Various	CRE (47)	CRE (26)	ND	26	ND	ND	ND	[32]
Chandigarh, Punjab	2008, 2012	Urine	CRE (222)NFGNB (309)	CRE (72)NFGNB (94)	NP	47	53	6	ND	[35]
Chandigarh, Punjab	2013–14	Blood, respiratory	AB (150)	AB (150)	NP	28	74	NP	NP	[30]
Srinagar, Jammu & Kashmir	2011–12	Various	CRE (38)NFGNB (62)	NA	ND	15	ND	ND	ND	[36]
New Delhi, Delhi	2011–12	Feces (surveillance)	CRE (91)	CRE (50)	2	39	ND	ND	18	[25]
New Delhi, Delhi	2011–13	Blood	CRE (93)	CRE (71)	2	61	6	NP	45	[26]
Lucknow, UP	2012	Various	NA	CRE (57)	22	57	2	12	13	[29]
Kanpur, UP	2014–16	Various	CRE &NFGNB (312)	CRE (144)NFGNB (159)	NP	178	52	5	30	[37]
Gurugram, Haryana	2017–18	Various	CRE (156)	CRE (127)	NP	54	NP	ND	49	[51]
South										
Bengaluru, Karnataka	2009	Various	CRE (51)	CRE (51)	ND	37	7	ND	ND	[38]
Bengaluru, Karnataka	2011–12	Various	GNB (69)	GNB (34)	ND	34	ND	ND	ND	[39]
Chennai, Tamil Nadu	2009	Various	CRE (141)	CRE (44)	ND	44	ND	ND	ND	[32]
Chennai, Tamil Nadu	2010	Various	CRE (111)	CRE (67)	ND	64	7	NP	2	[40]
Chennai, Tamil Nadu	2010	Various	PsA (61)	PsA (39)	ND	4	33	2	ND	[45]
Chennai, Tamil Nadu	2015–16	Various	CRE (105)NFGNB (23)	CPE (60)NFGNB (11)	NP	22	NP	3	21	[48]
Vellore, Tamil Nadu	2013–15	Blood	CRE (176)	CRE (165)	NP	80	21	NP	83	[30]
Vellore, Tamil Nadu	2015–16	Blood	CRKP (73)	CRKP (33)	NP	22	06	NP	15	[12]
East										
Kolkata, West Bengal	2007–11	Blood	CRE (15)	CRE (15)	ND	15	ND	ND	ND	[41]
Kolkata, West Bengal	2017–18	Blood	NA	CRE (123)NFGNB (31)	4	76	5	NP	86	[27]
Kolkata, West Bengal	2013–16	Blood	CRE (92)	CRE (88)	4	73	ND	NP	11	[28]
West										
Pune, Maharashtra	2011–13	Various	*Enterobacter* spp. (70)	*Enterobacter* spp. (48)	NP	19	29	ND	ND	[46]
Mumbai, Maharashtra	2012	Various	CRE (113)	CRE (111)	NP	106	4	NP	22	[42]
North-east										
Agartala, Tripura	2016–17	Blood	CRKP (26)	CRKP (12)	NP	12	NP	NP	NP	[33]

AB, *Acinetobacter baumannii*; CRE, carbapenem-resistant Enterobacterales; NFGNB, non-fermenter Gram-negative bacilli; ND, not done; NP, not present; NA, not available; P, present; PsA, *Pseudomonas aeruginosa*; UP, Uttar Pradesh.
[a] Confirmed by PCR.

pneumoniae isolates and rarely in *E. coli, P. aeruginosa, A. baumannii* and other members of the Enterobacterales family [27,29].

The most clinically relevant carbapenemase, NDM-1, was first identified in *K. pneumoniae* and *E. coli* from a Swedish patient previously hospitalized in New Delhi, India during 2009 [5]. Currently, 24 variants of NDM-1 have been identified globally, among which NDM-3, 4, 5, 6 and 7 are mainly reported in Asia, including India, but rarely in Africa and America [8,31,32]. The Asian continent, mainly China and India, is the hot-spot for NDM-1 producing Enterobacterales (58%) [17,31]. In association with the Indian subcontinent, NDM-1 producers were reported from other parts of the world, including Europe (16.8%), America (10.8%), Africa (10.8%) and Australia (1.6%) [5,6,9]. NDM-1 is carried by large conjugative plasmids which can cater up to fourteen antibiotic resistance genes and rapidly disseminate to other bacteria causing multidrug or pan-drug resistance [28,31,33,34]. NDM-1 has been reported from Pan-India with an isolation rate up to 90% (Table 1) [35–42]. In a surveillance study, NDM-1-producing CRE was noted as the major colonizer in intensive care unit (ICU) patients from New Delhi [25]. Exchange of the NDM-1 gene among unrelated bacterial isolates was observed in Enterobacterales and *A. baumannii,* facilitating difficult-to-control global dissemination of NDM-1 [8,31]. In addition to *K. pneumoniae* and *E. coli*, NDM-1 is rarely carried by other Gram-negative pathogens, including *Enterobacter cloacae*, *Citrobacter freundii* and *P. aeruginosa* isolates (Table 2). In the SENTRY Antimicrobial Surveillance Program during the 2006–07 period, CRE was detected in 2.7% (26/1,443) of the isolates collected from 14 hospitals in India. NDM-1 was detected among 58% (15/26) of the isolates predominantly in *E. coli* and *Klebsiella* spp. from New Delhi, Mumbai and Pune [43]. In 2009, the SMART study covering five continents showed 28% (66/235) of NDM-1 producers from eight sites from India [44]. Interestingly, Mukherjee et al. noted 6% (12/200) NDM-1 positivity among *K. pneumoniae* blood isolates in a neonatal sepsis study from north-east India [33].

In 1997, VIM-type (VIM-1) MBL, commonly associated with integron, was first isolated from *P. aeruginosa* in Verona, Italy [6]. Currently, 46 VIM-types have been assigned by the Lahey Clinic and VIM-2 is the most common type worldwide [8]. VIM-2 carbapenemase is carried by *Pseudomonas* spp., while VIM-1 is mainly found in *Enterobacterales* [9]. VIM-type carbapenemase, have been found in more than 17 countries, including India, and is prevalent in Africa and Europe [5]. In India VIM-positive CROs, especially *Pseudomonas* spp., have been mainly noted in northern and southern parts with an isolation rate up to about 80% (Table 1) [30,35,37,45]. Khajuria et al. reported the presence of VIM-2 and VIM-6 in 60% of *Enterobacter* spp. from west India [46]. Sharma et al. documented the presence of VIM-1 in 50% of CRAB isolates from north India [30]. During 2006–07, VIM-2, 5, 6 and 11 was detected in *Pseudomonas* spp. from multiple Indian cities by SENTRY Program [47]. This global study reported one novel variant (VIM-18) in *Pseudomonas* spp. from Kolkata, India [47]. Additionally, few researchers showed the presence of VIM in *E. coli, Klebsiella* spp., *Morganella* spp., *Providencia* spp., and *Alcaligenes* spp. in India (Table 2).

Table 2
Prevalence of "big five" carbapenemases among Enterobacterales and non-Enterobacterales in India.

Carbapenemase	Common organism	Rare organism	Ref.
Big-five genes			
KPC	*Klebsiella* spp.	*Escherichia coli*, *Acinetobacter* spp.	[27–29]
NDM-1	*Escherichia coli*, *Klebsiella* spp., *Acinetobacter* spp., *Pseudomonas* spp.	*Proteus* spp., *Enterobacter* spp., *Morganella* spp., *Providencia* spp., *Citrobacter* spp., *Stenotrophomonas* spp., *Alcaligenes* spp., *Serratia* spp.	[27,29,35,37,42,46,48,51]
VIM	*Pseudomonas* spp.	*Escherichia coli*, *Klebsiella* spp., *Enterobacter* spp., *Morganella* spp., *Providencia* spp., *Alcaligenes* spp.	[26,35,37,46,47]
IMP	*Pseudomonas* spp.	*Klebsiella* spp., *Acinetobacter* spp., *Enterobacter* spp., *Proteus* spp.	[35,37,48]
OXA-48-type	*Escherichia coli*, *Klebsiella* spp.	*Enterobacter* spp., *Acinetobacter* spp., *Proteus* spp.	[27,37,48,51]
Combinations	**Organism**		
OXA-48-type, NDM-1	*Escherichia coli*, *Klebsiella* spp., *Enterobacter* spp.		[26,27,42,48,51]
OXA-48-type, VIM	*Escherichia coli*		[26]
NDM, KPC	*Klebsiella* spp.		[29]
OXA-48-type, KPC	*Klebsiella* spp.		[29]
KPC, VIM	*Klebsiella* spp., *Pseudomonas* spp.		[27,29]
NDM, VIM	*Klebsiella* spp., *Escherichia coli*, *Pseudomonas* spp.		[27,42]
KPC, IMP	*Klebsiella* spp.		[29]
OXA-48-type, IMP	*Escherichia coli*		[29]
NDM, IMP	*Escherichia coli*		[29]
NDM-1, VIM, OXA-48-type	*Escherichia coli*		[26,29]

CRE; *Escherichia coli*, *Klebsiella* spp., *Enterobacter* spp., *Proteus* spp., *Citrobacter* spp., *Morganella* spp., *Providencia* spp. spp.
NFGNB; *Alcaligenes* spp., *Stenotrophomonas* spp., *Pseudomonas* spp., *Acinetobacter* spp., *Achromobacter* spp., *Elizabethkingia* spp.

The IMP-type (IMP-1) was first recognized in *P. aeruginosa* from Japan in 1988 [6]. Since then, 53 IMP-types have been identified among at least 26 species of Gram-negative bacteria especially in *Acinetobacter* and *Pseudomonas* and rarely in Enterobacterales from more than 24 countries/regions, importantly Japan, Taiwan, China, Indonesia and Singapore [5,9]. Indian researchers have rarely noted the presence of IMP in *Pseudomonas* spp. from northern and southern sites (Table 1). Other species like *Klebsiella* spp., *Acinetobacter* spp., *Enterobacter* spp. and *Proteus* spp. rarely carry IMP (Table 2).

The OXA-48 carbapenemase was first detected in a clinical isolate of *Klebsiella pneumoniae* from Istanbul, Turkey in 2001 [6]. To date, 10 variants of OXA-48-type are known among which OXA-48 is widespread among Enterobacterales followed by *A. baumannii* in several countries of the Middle East (Saudi Arabia and Israel), Africa (Libya, Egypt, Algeria, Morocco and South Africa), Asia (Russia, India, China and Taiwan) and South America (Argentina, Brazil and Colombia) [5]. In India, researchers have reported up to about 60% positivity of OXA-48-type commonly in *E. coli* and *Klebsiella* spp. (Table 1). OXA-181, the second most common, and OXA-232, the third most common member of OXA-48-type, has been increasingly reported in Asia, especially India [34,48,49]. During 2006–07, OXA-181 was first identified in Kolkata, India, followed by dispersion to other parts of Asia, and the United-States, United-Kingdom and South Africa [43,50]. In the SENTRY study, OXA-181 was detected from Kolkata, New Delhi and Mumbai among 42% (10/26) of CRE especially *Klebsiella* spp. [43]. In 2013, OXA-232 was first reported from a French patient with a history of travel to India followed by spread to Asia, Europe and the United-States [50]. Shankar et al. reported about 80% (60/74) of *K. pneumoniae* harboring OXA-232 from north and south India [49]. OXA-48-type carrying CROs are frequently co-harbor MBL, especially NDM-1 (Table 2) [27,34,48,51]. In neonatal sepsis cases, *K. pneumoniae* have been noted to co-harbor NDM-5 and OXA48-type from eastern India [28]. Although OXA-48-type is endemic in India, several researchers have not screened the prevalence of this carbapenemase (Table 1). Among CP-CROs, *E. coli* and *Klebsiella* spp. are common single- and multi-carbapenemase producers (Table 2).

4.2. Animal, food and environmental origin

Emergence of CROs in food animals, pharmaceutical and hospital effluents pose a monumental risk to public health, challenging food safety and food security [19]. From 2010 onwards, CP-CROs were frequently reported in China and Germany in livestock and seafood. Different variants of NDM including NDM-1, 4, 5, 7, 9 and 17 were detected among Enterobacterales in chickens, cows and pigs from China, the United-States and Algeria, while OXA-48-type was reported in food animals from Lebanon and Italy [10,11]. In Germany, VIM-1 carrying Enterobacterales was commonly reported in pigs and seafood during 2011–16 [10]. IMP-27 was reported among CPE in pigs in the United-States during 2015 [21]. CRE harboring NDM-1 and OXA-48 have also been sporadically reported in wildlife and companion animals mainly dogs and cats from China, Germany, Algeria and the United-States, while CPE with different IMP-variants have been noted in cats and silver gulls from Australia [10]. In India, a limited number of studies have been conducted to capture the scenario of CP-CRO from animal origin (Table 3). In 2012, the first report of one *E. coli* isolate carrying NDM-5 was registered from a milk sample of mastitis cattle in eastern India [52]. During 2014–16, one multicenter study first confirmed the presence of NDM-1 and 5 among eight *E. coli* isolates from piglets in north and north-east India [53]. On the other hand, existence of VIM and OXA-48 was reported in cattle and pigs from north India, respectively [23,54]. Of note, all of these are border states of India connecting Nepal, Bhutan, and Bangladesh (Fig. 1). This may cause the cross-border transmission of CP-CRO via the food-chain supply.

Few researchers around the globe have highlighted the presence of carbapenemases from food/environmental sources [18,55,56]. Environmental bacteria are intrinsically resistant to carbapenems and are found to carry carbapenemases which are transferred to clinically relevant bacteria. KPC (Brazil, Portugal and China), NDM-1 (China and Vietnam), VIM (Germany, Portugal, Switzerland and Tunisia) and OXA-48-type (Germany) have been sporadically registered from hospital effluents and river waters since 2005 [56]. Environmental studies have been rarely conducted

Table 3
Distribution of clinically important carbapenemases among animal-, food- and environment-derived isolates in India.

Region/Location	Period	Sample	CRO (*n*)	CP-CRO (*n*)	Carbapenemases (*n*)[a]					Ref.
					KPC	NDM	VIM	IMP	OXA-48-type	
Animal										
Kolkata, West Bengal	2012	Milk (mastitis cattle)	CREC (1)	CREC (1)	ND	1	ND	ND	ND	[52]
Bareilly, UP	2013–14	Feces (healthy & diarrheal calves)	CREC (81)	CREC (1)	NP	NP	01	NP	NP	[23]
Bareilly, UP Sitapur, UP Guwahati, AssamBarapani (Umiam) , Meghalaya	2014–16	Feces (Pig)	CREC (23)	CREC (8)	NP	8	NP	NP	NP	[54]
Bareilly, UP	2014–16	Feces (Pig)	CREC (27)	CREC (3)	NP	NP	NP	NP	3	[55]
Food/Environment										
Ujjain, UP	2008–10	Hospital effluent	CREC (6)	CREC (6)	NP	1	NP	NP	NP	[59]
New Delhi, Delhi	2010	Seepage & tap water	GNB (20)	GNB (11)	ND	11	ND	ND	ND	[57]
New Delhi, Delhi	2014	Hospital effluent, sewage treatment plant (STP), drain water, river water (Yamuna)	CRE (49)	CRE (>50%)	ND	>50%	ND	ND	ND	[58]
Hyderabad, Telangana	2016	Tap water, bore hole water, pharmaceutical effluents	CRE (25) NFGNB (26)	CRE (23) NFGNB (25)	7	10	5	5	22	[60]
Pune, Maharashtra	2014	Mutha river water	GNB	NA	P	P	NP	P	P	[53]
Mumbai, Maharashtra	2013–14	Fresh seafood	CRE (69)	CRE (2)	ND	2	ND	ND	ND	[61]
Mumbai, Maharashtra	2018	Fresh seafood	CRE (6)	CRE (6)	NP	6	NP	NP	NP	[62]

CRE, carbapenem-resistant Enterobacterales; CREC, carbapenem-resistant *Escherichia coli*; NFGNB, non-fermenter Gram-negative bacilli; ND, not done; NP, not present; NA, not available; P, present; UP, Uttar Pradesh.
[a] Confirmed by PCR.

in India (Table 3). NDM-1 among Enterobacterales and non-fermenters was noted in river water (Yamuna), sewage treatment plants, hospital effluents, seepage and tap water from north India [57–59]. In a point prevalence study from New Delhi, Walsh et al. detected NDM-1 for the first time from water and seepage samples among 11 bacterial species including *Shigella boydii* and *Vibrio cholerae* [57]. In addition to NDM-1, Lubbert et al. recorded OXA-48, KPC, IMP and VIM producing CROs from more than 95% of different environmental samples, including water sources and pharmaceutical effluents in south India during 2016 [60]. Both Singh et al. and Das UN et al. also reported the presence of NDM-1 in fresh seafood from the western part of India [61,62]. Indian pharmaceutical companies manufacture a large proportion of the world's antibiotics and hospitals also rampantly use antibiotics [2,4,55,57,60]. Pharmaceutical- and hospital-effluents containing antibiotic residues contaminate the environment, including the water bodies by selectively promoting superbugs such as CP-CROs. Food-producing animals, seafood and even humans are obscurely exposed to antibiotic pollution and superbugs.

5. Plasmids carrying carbapenemases

The existence of carbapenemase genes on plasmids promotes the rapid dissemination of carbapenem resistance across the same and different species, challenging the infection prevention practices in healthcare facilities. Different plasmid types have been reported in CROs, Including Incompatibility (Inc) types F, N, X, A/C, L/M, R, P, H, I and W around the globe among which IncF, A/C and X plasmids commonly carry carbapenemase genes [6,9,63]. Few Indian researchers have studied the plasmid types mediating carbapenem resistance in *E. coli* and *K. pneumoniae* (Table 4). IncF, narrow-host-range conjugative plasmids, are predominantly harbored by CRE around the globe, Including India. IncF, especially IncFII, FIIA and FIIK, disseminates NDM and OXA-48-type among *E. coli* and *K. pneumoniae* in South and Southeast Asia, while KPC is carried in the Western continent [21,28,33,34,48,63]. Other antibiotic resistance genes importantly ESBL (CTX-M, SHV), plasmid-mediated quinolone resistance (PMQR) (*qnr, oqx, aac (6′)-lb-c*), aminoglycoside resistance (*aadA2*), sulfonamide resistance (*sul1, sul2*) and trimethoprim resistance (*dfrA12, dfrA14*), have been found concurrently with carbapenemase genes on IncFII plasmid globally including India [28,33,34,63]. Conjugative IncX plasmid harbors the β-lactamases, including carbapenemases (mainly NDM-1 and KPC) [63]. One Indian study reported the occurrence of IncX3 plasmid in *E. coli* blood isolate carrying NDM-1 [48]. Notably, both IncFII and IncX3 plasmids have the potential to carry the colistin resistance gene (*mcr-1*) as evidenced by a Chinese study [64]. Additionally, IncFII and IncX3 plasmids have been isolated from animal sources in China and Algeria [63].

6. Carbapenem-resistant clones circulating in India

Diversity at genetic level defines most phenotypic variability in bacteria, especially geographic distribution and AMR. Multilocus sequence typing (MLST) has enhanced the diagnosis, treatment, epidemiological surveillance and outbreak detection of MDR Gram-negative infections, resulting in optimal antimicrobial and infection prevention stewardship. The linkage between clinically important carbapenemases, especially NDM, OXA-48-type and KPC, and specific sequence types (STs) have been established by several studies that provide useful information on the circulation of international and national clones of CRE [21,49,65]. Limited studies from India have shown the STs in CROs, especially *E. coli* and *K. pneumoniae* human-derived isolates (Table 5). In India, *E. coli* carrying NDM-1 belonging to ST2, ST10, ST44, ST88, ST131, ST471, ST501 and ST5053 have been registered, among which ST501 is the unique ST [44,54,62]. ST131 is a high-risk clone of extraintestinal pathogenic *E. coli* mediating large clonal expansion which is commonly associated with a global dispersion of KPC, NDM-1, and/or OXA-48-type [65]. ST131 causes infections mainly in humans; it has rarely been reported from other sources [11,21]. ST10 carrying KPC, VIM and/or OXA-48-type carbapenemase(s) are commonly reported to circulate in humans, animals and the environment globally [10,21]. ST88 carrying NDM-1 and/

Fig. 1. Presence of NDM-1 among human, animal, and environment interfaces in India.

or VIM circulate in humans globally, but the clone is restricted to animals in Europe [21]. Of note, ST10 and ST131 clones of *E. coli* are international HRCs transmitted from humans to animals and vice-versa (Table 5). In *K. pneumoniae*, 29 clones are found in India, among which ST29, ST347, ST572, ST3249, ST3418, ST3419, ST3420, ST3421 and ST3422 are novel clones carrying NDM-1

Table 4
Plasmid types associated with carbapenem resistance in India.

Inc type	Carbapenemase(s)	Source and country	Ref.
Escherichia coli			
A/C	NDM-1, KPC	Human: India, Iran, Kuwait, USA, Myanmar Environment: India	[21,48,57,63]
FIA-FIB	NDM-1, OXA-181	Human: India	[48]
FII	NDM-1	Human: China, Mexico, Myanmar, Vietnam Animal: China	[63]
FIIA	OXA-181, KPC-3	Human: India, Canada	[48,63]
FIIK	KPC	Human: USA	[63]
X3	NDM-1, OXA-181, KPC-3	Human: India, Kuwait, Lebanon, China, USA, Myanmar, Denmark, Korea, South Africa Cattle: Algeria, China	[21,48,63]
Klebsiella pneumoniae			
A/C	NDM-1, KPC, VIM-4	Human: China, Canada, Kuwait, Vietnam	[63]
FIA-FIB	NDM-1, KPC-2, OXA-48	Human: India, Istanbul, Australia, Canada, Ireland, South Africa	[21,48,63]
FIC	NDM-1	Human: India	[33]
FII	NDM-5, KPC-2, OXA-48	Human: India, Iran, Kuwait, UAE, Mexico, Istanbul, China, Poland, Italy, Romania, Ireland	[21,28,33,34,63]
FIIK	NDM-5, OXA-181, KPC-2	Human: India, Istanbul, China	[21,28,33]
ColKP3	OXA-181, OXA-232	Human: India, Egypt, Czeck Republic	[28,34]
X3	NDM-1	Human: India, China	[48,63]

Table 5
Global prevalence of carbapenemase(s) producing carbapenem-resistant Indian clones.

Sequence type (ST)	Carbapenemase(s)	Source and country	Ref.
Escherichia coli			
2	NDM-1	Human: India Cow: China	[10,44]
10	OXA-48-type, KPC, VIM	Human: UK Cattle: Tunisia, Egypt Pig: India Dog: France, South Korea Domestic goose: France	[10,21,55]
44	NDM-1	Human: India Animal: USA	[21,44]
88	NDM-1, VIM	Human: India Cattle: France Other animals: Germany, Spain	[10,44,65]
131	NDM, KPC, OXA-48-type	Human: USA, UK, Iran Swine: Algeria Poultry: Italy, Spain, Egypt Cat: France, Italy Seafood: India, Algeria Environment: Iran	[10,21,62,65]
297	VIM	Cattle: India	[23]
471	NDM-1	Human: India	[44]
501	NDM-1	Human: India	[44]
5053	OXA-48-type	Pigs: India	[55]
Klebsiella pneumoniae			
11	NDM, KPC, OXA-48-type	Human: USA, Tunisia, UK, China, India Dog: Italy	[21,34,44]
14	NDM-1, OXA-48-type	Human: India, USA; Tunisia, UK, Thailand, France	[10,34,44]
15	OXA-181 & 232	Human: India Dogs: Germany	[10,28,34]
16	OXA-181 & 232	Human: India	[34]
20	NDM-1	Human: India, China	[21,44]
29	NDM-1	Human: India, Taiwan, Saudi Arabia, Yemen	[33]
23	NDM-1, OXA-232	Human: India, China	[28,34]
43	KPC, OXA-48-type	Human: India, Iran	[34,44]
48	OXA-181, NDM-5	Human: India	[28]
101	NDM-1, OXA-48-type	Human: India, Tunisia, UK, Thailand Cat, dog: Italy	[21,44]
147	NDM-1, OXA-48-type (181 and 232)	Human: India, Italy, Greece, Tunisia, UK	[44,49]
231	NDM-1, OXA-48-type (181 and 232)	Human: India, UK, Thailand	[5,34,44]
258	KPC, OXA-48-type	Human: India, Greece, USA, Israel, UK	[11,44]
340	NDM-1	Human: India, UK, Iran Cat, dog: Italy	[21,44]
347	NDM-1	Human: India	[33]
391	NDM-1	Human: India	[44]
395	OXA-232	Human: India, Europe, Africa	[34]
570	OXA-232	Human: India	[34]
571	NDM-1	Human: India, China	[44]

(*continued on next page*)

Table 5 (*continued*)

Sequence type (ST)	Carbapenemase(s)	Source and country	Ref.
572	NDM-1	Human: India	[44]
1224	NDM-1	Human: China, India	[33]
2040	OXA-232	Human: India	[34]
2558	NDM-1	Human: India	[33]
3249	NDM-1	Human: India	[34]
3418	OXA-48-type	Human: India	[34]
3419	ND	Human: India	[34]
3420	ND	Human: India	[34]
3421	ND	Human: India	[34]
3422	ND	Human: India	[34]

ND, not detected.
Novel clones of India are marked in bold.
International high-risk clones are marked in italic.

[33,44,49]. Several international HRCs such as ST11, ST14, ST147 and ST231 are predominantly reported in India infecting humans and associated with rapid spread of NDM-1 and/or OXA-48-type, several outbreaks and high mortality rates [28,44,49,65]. Interestingly, ST11 is known as NDM-1 harboring clone in Asia, while the same ST circulates in KPC in Europe [34]. ST23 and ST231 pose a dual threat of virulence and antibiotic resistance and commonly colonize the human gut causing primary bacteremia and challenging patient management [49,65]. ST258 is another international HRC of *K. pneumoniae* transmitting KPC-2 and KPC-3 in North America, Europe and China [63]. In 2009, Lascols et al. showed the existence of ST258 carrying NDM-1 and OXA-48 in one *K. pneumoniae* isolate from India [44]. Of note, most STs of *K. pneumoniae* are largely restricted to humans all over the world except for ST11, ST15, ST101 and ST340, which have been detected from companion animals in Europe (Table 5). Most of these international HRCs of *E. coli* and *K. pneumoniae* harbor IncFII and FIIK plasmids carrying multiple antibiotic resistance determinants including the "*big five*" carbapenemases, while IncX3 commonly carrying NDM-1 are noted among ST10 and ST14 especially in Asia [21,63].

7. AMR policies in India and their effectiveness

Since the emergence of NDM-1 in 2010, the issue of AMR has been prioritized in India. After that, the Food Safety and Standards Authority of India (FSSAI) published the national policy for containment of AMR in 2011 focusing on antibiotic use in seafood and food animals, prohibiting non-therapeutic use of antibiotics in animals and guiding on proper food labeling [15]. In collaboration with the NCDC, the Indian Council of Medical Research (ICMR) launched the AMR Surveillance and Research Network (AMRSN) in 2013 to understand the resistance among nosocomial pathogens for developing new diagnostics and candidate drug molecules [4]. In 2015, the WHO instituted the Global AMR Surveillance System (GLASS) which was implemented by 78 countries by July 2019, including India, for collecting epidemiological, clinical, and population-based AMR data to reduce the global burden of AMR [16]. The National Action Plan for containment of AMR (NAP-AMR) was launched in 2017 to promote investment in research [4]. AMR impacts the achievement of several sustainable development goals (SDGs) by 2030, especially health and wellbeing, food security, poverty reduction, environment and economic growth. The Food and Agriculture Organization of the United Nations (FAO), the World Organization of Animal Health (OIE) and the WHO, as well as the UN Environment Program (UNEP) jointly recommended to include two indicators for AMR as part of SDG 3 (good health and wellbeing) in 2021, namely frequency of BSI cases by specific antibiotic-resistant bacteria and availability of essential medicines in healthcare facilities [66]. Antimicrobial stewardship (AMS) and infection prevention stewardship (IPS) are pivotal for curbing AMR issue in the human–animal–environment interface [19]. In addition to Europe, the United-Kingdom and the United-States, the successful implementation of AMS programs in several countries of South and South-East Asia has resulted in a reduction of antimicrobial use, AMR, mortality, length of hospitalization and healthcare cost [13,14,24,67]. Implementation, monitoring, and assessment of these AMR policies remain challenging issues for lower-middle income countries (LMICs) like India [2,4,66,67]. Despite these policies, a lack of coordination among human–animal–environment domains continues to magnify the burden of carbapenem resistance in India.

8. Conclusion

Overall, carbapenem resistance surveillance is predominantly focused on human populations globally, including in India. Researchers are reluctant to consider the animal populations where carbapenems are not used. Transmission of these superbugs through food-producing animals compromises the food chains challenging the import and export services globally. Hence, human-derived CROs pose a massive risk for animal health and also as an environmental pollutant. For tracking and mapping the transmission dynamics of CP-CROs, it is highly recommended to analyze this global issue using the one health approach in India, where the burden of these superbugs is critically significant. Cross-border transmission of these superbugs, particularly the international HRCs, poses the threat of outbreak and rapid spread of multiple antibiotic resistance genes, importantly *mcr*, declining the efficacy of colistin which is being used both in human and animal infections. Circulation of CP-CRO especially in food animals among the Indian border states, emphasizes strict surveillance on import and export of animals. European countries have significantly reduced the emergence of AMR, including carbapenem resistance following strict one health concept. High versatile hydrolyzing activities of different carbapenemase genes demand in-depth epidemiological surveillance at a national level to curb the case fatality by initiating effective antibiotic therapy. India is the endemic country for NDM-1 producing CROs commonly circulating in humans followed by animals and environment origins. NDM-1 might be used as marker to track CP-CROs in human–animal–environment interfaces and to map carbapenem resistance in India for effective containment of these pathogens. Prescient execution of the one health approach at national level is urgently required for strengthening the antimicrobial and infection prevention stewardships throughout the country.

Disclosure of interest

The authors declare that they have no known competing financial or personal relationships that could be viewed as influencing the work reported in this paper.

Authors' contribution

SD conceived the idea and prepared the final manuscript.

Human and animal rights

The authors declare that the work described has not involved experimentation on humans or animals.

Informed consent and patient details

The authors declare that the work described does not involve patients or volunteers.

Funding

This work did not receive any grant from funding agencies in the public, commercial, or not-for-profit sectors.

References

[1] Murray CJ, Ikuta KS, Sharara F, Swetschinski L, Aguilar GS, Gray A, et al. Global burden of bacterial antimicrobial resistance in 2019: a systematic analysis. Lancet 2022;399:629–55. https://doi.org/10.1016/S0140-6736(21)02724-0.
[2] Laxminarayan R, Matsoso P, Pant S, Brower C, Rottingen JA, Klugman K, et al. Access to effective antimicrobials: a worldwide challenge. Lancet 2016;387:168e75. https://doi.org/10.1016/S0140-6736(15)00474-2.
[3] Taneja N, Sharma M. Antimicrobial resistance in the environment: the Indian scenario. Indian J Med Res 2019;149:119–28. https://doi.org/10.4103/ijmr.IJMR_331_18.
[4] Gandra S, Joshi J, Trett A, Lamkang AS, Laxminarayan R. Scoping report on AMR in India. Washington DC: Center for Disease Dynamics, Economics and Policy; 2017. https://cddep.org/publications/scoping-report-antimicrobial-resistance-india/ (accessed 10 Sep 2022).
[5] Nordmann P, Poirel L. The difficult-to-control spread of carbapenemase producers among *Enterobacteriaceae* worldwide. Clin Microbiol Infect 2014;20:821–30. https://doi.org/10.1111/1469-0691.12719.
[6] Temkin E, Adler A, Lerner A. Carbapenem-resistant *Enterobacterales*: biology, epidemiology, and management. Ann N Y Acad Sci 2014;1323:22–42. https://doi.org/10.1111/nyas.12537.
[7] Saltin JM, Chen L, Patel G, Gomez-Simmonds A, Weston G, Kim AC, et al. Multicenter clinical and molecular epidemiological analysis of bacteremia due to carbapenem-resistant *Enterobacteriaceae* (CRE) in the CRE epicenter of the United States e02349-16. Antimicrob Agents Chemother 2016;61. https://doi.org/10.1128/AAC.02349-16.
[8] Queenan AM, Bush K. Carbapenemases: the versatile β-lactamases. Clin Microbiol Rev 2007;20:440–58. https://doi.org/10.1128/CMR.00001-07.
[9] Codjoe FS, Donkor ES. Carbapenem resistance: a review. Med Sci 2018;6:1–28. https://doi.org/10.3390/medsci6010001.
[10] Kock R, Daniels-Haardt I, Becker K, Mellmann A, Friedrich AW, Mevius D, et al. Carbapenem-resistant *Enterobacteriaceae* in wildlife, food-producing and companion animals– a systematic review. Clin Microbiol Infect 2018;24:1241–50. https://doi.org/10.1016/j.cmi.2018.04.004.
[11] Madec JY, Haenni M, Nordmann P, Poirel L. Extended-spectrum β-lactamase/AmpC- and carbapenemase-producing *Enterobacteriaceae* in animals: a threat for humans. Clin Microbiol Infect 2017;23:826–33. https://doi.org/10.1016/j.cmi.2017.01.013.
[12] Veeraraghavan B, Shankar C, Karunasree S, Kumari S, Ravi R, Ralph R. Carbapenem resistant *Klebsiella pneumoniae* isolated from bloodstream infection: Indian experience. Pathogens Global Health 2017;111:240–6. https://doi.org/10.1080/20477724.2017.1340128.
[13] ECDC/EFSA/EMA (European Centre for Disease Prevention and Control-European Food Safety Authority- European Medicines Agency), 2017.ECDC/EFSA/EMA second joint report on the integrated analysis of the consumption of antimicrobial agents and occurrence of antimicrobial resistance in bacteria from humans and food-producing animals. www.ema.europa.eu/docs/en_GB/document_library/Report/2017/07/WC500232336.pdf (accessed 10 Sep 2022).
[14] AVMA (American Veterinary Medical Association). One Health Initiative Task Force: final report. One Health: a new professional imperative. AVMA; 2008. https://www.avma.org/resources-tools/reports/one-health-ohitf-final-report-2008 (accessed 10 Sep 2022).
[15] Antibiotic use and resistance in food animals: current policy and recommendations. Center for Disease Dynamics, Economics & Policy (CDDEP). Washington DC & New Delhi. 2016. https://cddep.org/publications/antibiotic_use_and_resistance_food_animals_current_policy_and_recommendations/ (accessed 10 Sep 2022).
[16] Global antimicrobial resistance surveillance system (GLASS) report: early implementation 2020. Geneva: World Health Organization; 2020, https://www.who.int/publications/i/item/9789240005587 (accessed 10 Sep 2022).
[17] Queenan K, Häsler B, Rushton J. A One Health approach to antimicrobial resistance surveillance: is there a business case for it? Int J Antimicrob Agents 2016;48:422–7. https://doi.org/10.1016/j.ijantimicag.2016.06.014.
[18] Ashbolt NJ, Amezquita A, Backhaus T, Borriello P, Brandt KK, Collignon P, et al. Human health risk assessment (HHRA) for environmental development and transfer of antibiotic resistance. Environ Health Perspect 2013;121:993–1001. https://doi.org/10.1289/ehp.1206316.
[19] Collignon PC, Conly JM, Andremont A. World Health Organization Advisory Group, Bogota Meeting on Integrated Surveillance of Antimicrobial Resistance (WHO-AGISAR)., et al. World health organization ranking of antimicrobials according to their importance in human medicine: A critical step for developing risk management strategies to control antimicrobial resistance from food animal production. Clin Infect Dis 2016; 63: 1087–93, https://doi.org/10.1093/cid/ciw475.
[20] Mahalmani VM, Sharma P, Prakash A, Medhi B. Positive list of antibiotics and food products: current perspective in India and across the globe. Indian J Pharmacol 2019;51:231–5. https://doi.org/10.4103/ijp.IJP_548_19.
[21] Dandachi I, Chaddad A, Hanna J, Matta J, Daoud Z. Understanding the epidemiology of multi-drug resistant Gram-negative bacilli in the Middle East using a one health approach. Front Microbiol 2019. https://doi.org/10.3389/fmicb.2019.01941.
[22] Sivalingam P, Poté J, Prabakar K. Environmental prevalence of carbapenem resistance *Enterobacteriaceae* (CRE) in a tropical ecosystem in India: human health perspectives and future directives. Pathogens 2019;8:174. https://doi.org/10.3390/pathogens8040174.
[23] Murugan MS, Sinha DK, Vinodh Kumar OR, Yadav AK, Pruthvishree BS, Vadhana P, et al. Epidemiology of carbapenemresistant Escherichia coli and first report of blaVIM carbapenemases gene in calves from India. Epidemiol Infect 2019;147:e159. https://doi.org/10.1017/S0950268819000463.
[24] Destoumieux-Garzón D, Mavingui P, Boetsch G, Boissier J, Darriet F, Duboz P, et al. The One Health Concept: 10 Years old and a long road ahead. Front Vet Sci 2018;5:14. https://doi.org/10.3389/fvets.2018.00014.
[25] Mittal G, Gaind R, Kumar D, Kaushik G, Gupta KB, Verma PK, et al. Risk factors for fecal carriage of carbapenemase producing *Enterobacteriaceae* among intensive care unit patients from a tertiary care center in India. BMC Microbiol 2016;16:138. https://doi.org/10.1186/s12866-016-0763-y.
[26] Mohanty S, Mittal G, Gaind R. Identification of carbapenemase-mediated resistance among *Enterobacteriaceae* bloodstream isolates: A molecular study from India. Indian J Med Microbiol 2017;35:421–5. https://doi.org/10.4103/ijmm.IJMM_16_386.
[27] Das S, Roy S, Roy S, Goel G, Walia K, Mukherjee S, et al. Rapid and economical detection of eight carbapenem-resistance genes in *Enterobacteriaceae*, *Pseudomonas* spp., and *Acinetobacter* spp. directly from positive blood cultures using an internally controlled multiplex-PCR assay. Infect Control Hospital Epidemiol 2019;40:737–9. https://doi.org/10.1017/ice.2019.79.
[28] Naha S, Sands K, Mukherjee S, Saha B, Dutta S, Basu S. OXA-181-like carbapenemases in *Klebsiella pneumoniae* ST14, ST15, ST23, ST48, and ST231 from septicemic neonates: coexistence with NDM-5, resistome, transmissibility, and genome diversity e01156-20. mSphere 2021;6. https://doi.org/10.1128/mSphere.01156-20.
[29] Rahman M, Shukla SK, Prasad KN, Ovejero CM, Pati BK, Tripathi A, et al. Prevalence and molecular characterization of New-Delhi metallo-β-lactamases NDM-1, NDM-5, NDM-6 and NDM-7 in multidrug-resistant *Enterobacteroceae* from India. Int J Antimicrob Agents 2014;44:30–7. https://doi.org/10.1016/j.ijantimicag.2014.03.003.
[30] Sharma A, Bakthavatchalam YD, Gopi R, Anandan S, Verghese VP, Veeraraghavan B. Mechanisms of carbapenem resistance in *K. pneumoniae* and *E. coli* from bloodstream infections in India. J Infect Dis Ther 2016;4:293. https://doi.org/10.4172/2332-0877.1000293.
[31] Wu W, Feng Y, Tang G, Qiao F, McNally A, Zong Z. NDM metallo-β-lactamases and their bacterial producers in health care settings e00115–18. Clin Microbiol Rev 2019;32. https://doi.org/10.1128/CMR.00115-18.
[32] Kumarasamy KK, Toleman MA, Walsh TR, Bagaria J, Butt F, Balakrishnan R, et al. Emergence of a new antibiotic resistance mechanism in India, Pakistan, and the UK: a molecular, biological, and epidemiological study. Lancet Infect Dis 2010;10:597–602. https://doi.org/10.1016/S1473-3099(10)70143-2.
[33] Mukherjee S, Bhattacharjee A, Naha S, Majumdar T, Debbarma SK, Kaur H, et al. Molecular characterization of NDM-1-producing *Klebsiella pneumoniae* ST29, ST347, ST1224, and ST2558 causing sepsis in neonates in a tertiary care hospital of North-East India. Infect Genet Evol 2019;69:166–75. https://doi.org/10.1016/j.meegid.2019.01.024.
[34] Shankar C, Jacob JJ, Sugumar SG, Natarajan L, Rodrigues C, Mathur P, et al. Distinctive mobile genetic elements observed in the clonal expansion of carbapenem-resistant *Klebsiella pneumoniae* in India. Microb Drug Resist 2021;27:1096–104. https://doi.org/10.1089/mdr.2020.0316.

[35] Mohan B, Hallur V, Singh G, Kaur HS, Suma B, Taneja N. Occurrence of *bla*NDM-1 & absence of *bla*KPC genes encoding carbapenem resistance in uropathogens from a tertiary care centre from north India. Indian J Med Res 2015;142:336–43. https://doi.org/10.4103/0971-5916.166601.
[36] Fomda BA, Khan A, Zahoor D. NDM-1 (New Delhi metallo beta-lactamase-1) producing Gram-negative bacilli: Emergence and clinical implications. Indian J Med Res 2014; 140:672-8, PMC4311323.
[37] Garg A, Garg J, Kumar S, Bhattacharya A, Agarwal S, Upadhyay GC. Molecular epidemiology and therapeutic options of carbapenem-resistant Gram-negative bacteria. Indian J Med Res 2014;149:285–9. https://doi.org/10.4103/ijmr.IJMR_36_18.
[38] Nagaraj S, Chandran SP, Shamanna P, Macaden R. Carbapenem resistance among *Escherichia coli* and *Klebsiella pneumoniae* in a tertiary care hospital in south India. Indian J Med Microbiol 2012;30:93–5. https://doi.org/10.4103/0255-0857.93054.
[39] Shenoy A, Jyothi EK, Ravikumar R. Phenotypic identification & molecular detection of $bla_{\mathrm{NDM-1}}$ gene in multidrug resistant Gram-negative bacilli in a tertiary care centre. Indian J Med Res 2014; 139: 625-31, PMC4078503.
[40] Mariappan S, Sekar U, Kamalanathan A. Carbapenemase-producing *Enterobacteriaceae*: risk factors for infection and impact of resistance on outcomes. Int J App Basic Med Res 2017;7:32–9. https://doi.org/10.4103/2229-516X.198520.
[41] Datta S, Roy S, Chatterjee S, Saha A, Sen B, Pal T, et al. A Five-Year Experience of carbapenem resistance in *Enterobacteriaceae* causing neonatal septicaemia: predominance of NDM-1. PLoS ONE 2014;9(11):e112101. https://doi.org/10.1371/journal.pone.0112101.
[42] Kazi M, Drego L, Nikam C, Ajbani K, Soman R, Shetty A, et al. Molecular characterization of carbapenem-resistant *Enterobacteriaceae* at a tertiary care laboratory in Mumbai. Eur J Clin Microbiol Infect Dis 2015;34:467–72. https://doi.org/10.1007/s10096-014-2249-x.
[43] Castanheira M, Deshpande LM, Mathai D, Bell JM, Jones RN, Mendes RE. Early dissemination of NDM-1- and OXA-181-producing *Enterobacteriaceae* in Indian hospitals: report from the SENTRY antimicrobial surveillance program, 2006–2007. Antimicrob Agents Chemother 2011;55:1274–8. https://doi.org/10.1128/AAC.01497-10.
[44] Lascols C, Hackel M, Marshall SH, Hujer AM, Bouchillon S, Badal R, et al. Increasing prevalence and dissemination of NDM-1 metallo-β-lactamase in India: data from the SMART study. J Antimicrob Chemother 2009;66:1992–7. https://doi.org/10.1093/jac/dkr240.
[45] Shanthi M, Sekar U, Kamalanathan A, Sekar B. Detection of New Delhi metallo beta lactamase-1 (NDM-1) carbapenemase in *Pseudomonas aeruginosa* in a single centre in southern India. Indian J Med Res 2014; 140: 546-50, PMC4277142.
[46] Khajuria A, Praharaj AK, Kumar M, Grover N. Carbapenem resistance among *Enterobacter* Species in a tertiary care hospital in central India. Chemother Res Pract 2014:972646. https://doi.org/10.1155/2014/972646.
[47] Castanheira M, Bell JM, Turnidge JD, Mathai D, Jones RN. Carbapenem resistance among *Pseudomonas aeruginosa* strains from India: evidence for nationwide endemicity of multiple metallo-β-lactamase clones (VIM-2, -5, -6, and -11 and the newly characterized VIM-18). Antimicrob Agents Chemother 2009;53:1225–7. https://doi.org/10.1128/AAC.01011-08.
[48] Manohar P, Sebastian L, Lopes BS, Nachimuthu R. Dissemination of carbapenem resistance and plasmids encoding carbapenemases in Gram-negative bacteria isolated in India. JAC Antimicrob Resist 2021;3. https://doi.org/10.1093/jacamr/dlab015.
[49] Shankar C, Mathur P, Venkatesan M, Pragasam AK, Anandan S, Khurana S, et al. Rapidly disseminating blaOXA-232 carrying *Klebsiella pneumoniae* belonging to ST231 in India: multiple and varied mobile genetic elements. BMC Microbiol 2019;19:137. https://doi.org/10.1186/s12866-019-1513-8.
[50] Pitout JDD, Peirano G, Kock MM, Strydom KA, Matsumura Y. The global ascendency of OXA-48-type carbapenemases e00102-19. Clin Microbiol Rev 2019;33. https://doi.org/10.1128/CMR.00102-19.
[51] Jaggi N, Chatterjee N, Singh V, Giri SK, Dwivedi P, Panwar R, et al. Carbapenem resistance in *Escherichia coli* and *Klebsiella pneumoniae* among Indian and international patients in north India. Acta Microbiol Immunol Hung 2019;66:367–76. https://doi.org/10.1556/030.66.2019.020.
[52] Ghatak S, Singha A, Sen A, Guha C, Ahuja A, Bhattacharjee U, et al. Detection of New Delhi metallo-β-lactamase and extended-spectrum-β-actamase genes in *Escherichia coli* isolated from mastitic milk samples. Transbound Emer Dis 2013;60:385–9. https://doi.org/10.1111/tbed.12119.
[53] Marathe NP, Pal C, Gaikwad SS, Jonsson V, Kristiansson E, Larsson DGJ. Untreated urban waste contaminates Indian river sediments with resistance genes to last resort antibiotics. Water Res 2017;124:388–97. https://doi.org/10.1016/j.watres.2017.07.060.
[54] Pruthvishree BS, Vinodh Kumar OR, Sinha DK, Malik YPS, Dubal Z, Desingu PA, et al. Spatial molecular epidemiology of carbapenem resistant and New Delhi Metallo Beta-lactamase (blaNDM) producing *Escherichia coli* strains from farmed piglets of India. J App Microbiol 2017;122:1537–46. https://doi.org/10.1111/jam.13455.
[55] Nirupama KR, Kumar V, Pruthvishree BS, Sinha DK, Murugan MS, Krishnaswamy N, et al. Molecular characterisation of $\mathrm{bla}_{\mathrm{OXA-48}}$ carbapenemase-, extended-spectrum β-lactamase- and Shiga toxin-producing *Escherichia coli* isolated from farm piglets in India. J Glob Antimicrob Resist 2018;13:201–5. https://doi.org/10.1016/j.jgar.2018.01.007.
[56] Woodford N, Wareham DW, Guerra B, Teale C. Carbapenemase-producing *Enterobacteriaceae* and non-*Enterobacteriaceae* from animals and the environment: an emerging public health risk of our own making? J Antimicrob Chemother 2014;69:287–91. https://doi.org/10.1093/jac/dkt392.
[57] Walsh TR, Weeks J, Livermore DM, Toleman MA. Dissemination of NDM-1 positive bacteria in the New Delhi environment and its implications for human health: an environmental point prevalence study. Lancet Infect Dis 2011;11:355–62. https://doi.org/10.1016/S1473-3099(11)70059-7.
[58] Lamba M, Gupta S, Shukla R, Graham DW, Sreekrishnan TR, Ahammad SZ. Carbapenem resistance exposures via wastewaters across New Delhi. Environ Int 2011;119:302–8. https://doi.org/10.1016/j.envint.2018.07.004.
[59] Chandran SP, Diwan V, Tamhankar AJ, Joseph BV, Rosales-Klintz S, Mundayoor S, et al. Detection of carbapenem resistance genes and cephalosporin, and quinolone resistance genes along with oqxAB gene in *Escherichia coli* in hospital wastewater: a matter of concern. J Appl Microbiol 2014;117:984–95. https://doi.org/10.1111/jam.12591.
[60] Lübbert C, Baars C, Dayakar A, Lippmann N, Rodloff AC, Kinzig M, et al. Environmental pollution with antimicrobial agents from bulk drug manufacturing industries in Hyderabad, South India, is associated with dissemination of extended-spectrum beta-lactamase and carbapenemase-producing pathogens. Infect 2017;45:479–91. https://doi.org/10.1007/s15010-017-1007-2.
[61] Singh AS, Lekshmi M, Prakasan S, Nayak BB, Kumar S. Multiple antibiotic-resistant, extended spectrum-β-lactamase (ESBL)-producing enterobacteria in fresh seafood. Microorganisms 2017;5:53. https://doi.org/10.3390/microorganisms5030053.
[62] Das UN, Singh AS, Lekshmi M, Nayak BB, Kumar S. Characterization of blaNDM-harboring, multidrug-resistant *Enterobacteriaceae* isolated from seafood. Environ Sci Pollut Res Int 2019;26:2455–63. https://doi.org/10.1007/s11356-018-3759-3.
[63] Kopotsa K, Sekyere JO, Mbelle NM. Plasmid evolution in carbapenemase-producing *Enterobacteriaceae*: a review. Ann NY Acad Sci 2019;1457:61–91. https://doi.org/10.1111/nyas.14223.
[64] Zhang Y, Liao K, Gao H, Wang Q, Wang X, Li H, et al. Decreased fitness and virulence in ST10 *Escherichia coli* harboring *bla*NDM-5 and *mcr-1* against a ST4981 strain with *bla*NDM-5. Front Cell Infect Microbiol 2017;7:242. https://doi.org/10.3389/fcimb.2017.00242.
[65] Ewers C, Bethe A, Semmler T, Guenther S, Wieler LH. Extended-spectrum β-lactamase-producing and AmpC-producing *Escherichia coli* from livestock and companion animals, and their putative impact on public health: a global perspective. Clin Microbiol Infect 2012;18:646–55. https://doi.org/10.1111/j.1469-0691.2012.03850.x.
[66] Antimicrobial Resistance Multi-Partner Trust Fund annual report 2021. Geneva: World Health Organization, Food and Agriculture Organization of the United Nations, United Nations Environment Programme and World Organisation for Animal Health; 2022. https://www.who.int/publications/i/item/9789240051362, (accessed 10 Sep 2022).
[67] Huttner B, Harbarth S, Nathwani D, ESCMID Study Group for Antibiotic Policies (ESGAP). Success stories of implementation of antimicrobial stewardship: A narrative review. Clin Microbiol Infect 2014;20:954–62. https://doi.org/10.1111/1469-0691.12803.